Mobile Banking Financial Inclusion and Cybersecurity Risk in India

Entrepreneurial Finance | Term Paper | Group 31
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Abstract

Digital financial services can expand access to payments and savings, while safe use remains essential. Using the World Bank Global Findex 2025 India microdata, this study analyzes 3,000 adult respondents interviewed in 2024. Survey-weighted account ownership was 89.0% and 48.5% made or received a digital payment. Among mobile-phone owners, 13.1% of digital-payment users reported an unsolicited call or message asking for money, compared with 4.3% of non-users. A weighted exploratory logistic model yields an adjusted odds ratio of 3.45 for payment use; its uncertainty estimates do not incorporate the full survey design. Among phone owners 57.7% reported a device lock. These associations are not causal and unsolicited requests are not necessarily financial fraud. The findings motivate a proposed accessible safety and reporting service, whose impact and commercial viability require testing.

Keywords: financial inclusion, digital payments, mobile banking, cybersecurity, fraud, entrepreneurship, India

1 Introduction

A financial account can lower the friction of receiving wages, storing savings and paying suppliers. Mobile banking and digital payments extend those services beyond the branch, potentially helping small merchants and individuals transact at lower time and travel cost. However, ownership of an account is only the first stage of inclusion: a person must also be able to access, understand and safely use the service. Fraudulent calls, requests for credentials and misleading payment instructions can undermine confidence and impose losses.

India is a useful setting because account ownership is high while access to devices and use of digital payments remain uneven. This paper asks how inclusion and digital safety fit together, which groups may require additional support, and what business opportunity follows. The focus is the adult population, with relevance to potential entrepreneurs and small merchants; the Findex data do not identify a representative sample of businesses.

The research questions are: (1) How do account ownership and digital payments differ? (2) Are digital-payment users more likely to report unsolicited money requests? (3) Does device locking appear associated with sending money in response? (4) Which demographic groups show differences in adoption and device security? We use weighted descriptive statistics and an exploratory adjusted model. The cross-sectional data cannot establish causal effects or the sequence of exposure and adoption.

2 Background and related evidence

2.1 Inclusion as access and use

The World Bank defines account ownership to include accounts at financial institutions or use of a mobile money service. Digital payment measures are narrower than ownership and depend on the exact question: making a digital payment is not necessarily the same as making or receiving one. Phone ownership, smartphone access, connectivity, language, skills and trust can each influence whether a person uses an account digitally. A gender gap in payment use therefore cannot be attributed to cybersecurity alone.

The Global Findex 2025 combines financial inclusion indicators with questions about digital connectivity and safety. Its India survey was conducted in 2024 and the public India catalogue describes 3,000 individual records and 199 variables [1, 2]. The survey is cross-sectional; its 2025 name refers to the database release, not the survey year.

2.2 Digital safety and consumer trust

A phone password helps protect access to a device, but it cannot by itself prevent a user from authorizing a deceptive payment. Fraud exposure, sending money to a scammer, financial loss, and perceptions of safety are distinct outcomes. The connectivity questionnaire includes questions on phone PIN or password, receipt of a scam call or text, and sending money to a scammer [2]. A scam call or text is not necessarily connected to banking, and a reported exposure is not proof of a financial loss.

The Reserve Bank of India advises customers not to share passwords, PINs, one-time passwords, CVV or UPI PINs [3]. Practical protection also depends on providers offering understandable warnings, effective reporting and accessible redress. An app that simply teaches users to be careful cannot guarantee security or shift all responsibility for fraud to them.

3 Data and method

This paper analyzes the World Bank India Global Findex 2025 microdata, version IND_2024_FINDEX_v02_M, with 3,000 respondents interviewed in 2024 [2]. The file includes a survey weight but no published strata or primary sampling unit identifiers. Population shares use wgt; all sample sizes are unweighted. For binary derived indicators, 0 includes no, do not know and refusal as coded by the World Bank. For raw connectivity questions, only yes and no responses are included; don’t know, refusal and missing values are excluded. The survey questions on phone locks and unsolicited calls or messages were asked of phone owners, and the sending-money question follows reported exposure. We use published aggregates [4–6] as context.

We calculate weighted proportions as the sum of weights for positive responses divided by the sum of weights for eligible valid responses. Differences in percentages are descriptive. A weighted logistic regression of exposure includes digital payment use, gender, age, education, income quintile, workforce status and urbanicity. Income quintile is modeled as a linear ordinal term. Odds ratios describe conditional associations, not risk ratios or causation. Robust standard errors treat respondents as independent; because the released file lacks survey design identifiers, intervals and tests are exploratory and may understate or overstate uncertainty. No formal design-based significance claim is made.

4 Findings

4.1 Account access has expanded

The World Bank reports that 89.0% of Indian adults owned an account in 2024, compared with 77.5% in 2021 and 53.1% in 2014 [5]. The 2024 world figure was 78.7%. India’s 2024 account ownership was thus 10.3 percentage points above the world figure; the comparison is descriptive and does not measure frequency, quality or safety of account use. The Indian share rose 35.9 percentage points between 2014 and 2024, although the observations do not explain the contribution of any particular policy or technology.

Indicator	Women	Men	Difference
Account ownership	89.2%	88.8%	0.4 pp women higher
Digital payment use	36%	54%	18 pp men higher
Own a mobile phone	54.3%	78.4%	24.1 pp men higher
Own a smartphone	31.8%	52.0%	20.2 pp men higher
Internet use in past 30 days	36.3%	55.0%	18.7 pp men higher

Table 1. India, 2024: selected indicators among adults aged 15+. Account ownership is from [5]; digital payment use from [4]; connectivity from [6]. The payment indicator covers specified digital payment methods; it should not be equated with every broader measure of making or receiving payments. Differences use displayed values and can reflect rounding. These are separate indicators, not a joint distribution.

4.2 Ownership does not imply equal digital use

Women and men had almost identical account ownership, yet the dashboard reports an 18 percentage point difference in digital payment use [4]. The connectivity brief reports a 24.1 point gap in mobile phone ownership and an 18.7 point gap in recent internet use [6]. These patterns are consistent with barriers between opening an account and using digital services. They do not prove that a lack of a phone causes the payment gap, because these statistics do not show how the indicators overlap within individuals and do not account for other differences.

Income differences also matter. The dashboard reports account ownership of 86% among the poorest 40% and 91% among the wealthiest 60%, while digital payment use was 43% and 56%, respectively [4]. The displayed gaps are 5 and 13 percentage points. The figures suggest that a product designed only for confident smartphone users could miss a substantial part of the intended market.

4.3 What the available evidence can and cannot say about risk

Among phone owners, the survey records 219 reports of an unsolicited call or message asking for money, a weighted 9.4% of 2,145 phone owners. Of the 219, 45 reported sending money, a weighted 21.0% of those reporting such a request. These are self-reports, not verified losses, and the question does not establish that every request was fraudulent or connected to banking. The follow-up group is small and selected, so subgroup comparisons require caution.

5 Respondent level analysis and results

5.1 Adoption and exposure

Weighted estimates show that 89.0% of adults had an account and 48.5% made or received a digital payment (n = 3,000). Among 2,145 phone owners, 13.1% of digital-payment users (n = 1,283; 173 positive reports) reported receiving an unsolicited request for money, compared with 4.3% of non-users (n = 862; 46 positive reports), a difference of 8.8 percentage points. Digital payment use can include receiving money, while exposure is a call or text from an unknown person asking for money in the preceding 12 months; neither measure identifies a banking scam.

5.2 Device security and demographic differences

Among eligible phone owners, 57.7% had a device lock (n = 2,141; four uncertain responses excluded). The share was 67.4% for digital-payment users and 44.6% for non-users. A device lock protects access to a phone but may not prevent deceptive requests. Of the 219 people reporting an unsolicited request, 170 reported a lock and 49 did not. The weighted share reporting sending money was 25.0% and 8.9%, respectively (39 and 6 positive responses). The small, selected sample, potential confounding and unknown timing prevent any conclusion that locking increases or reduces the likelihood of sending money.

Among women and men, weighted account ownership was 89.2% and 88.8%, while making or receiving a digital payment was 42.7% and 54.1%, respectively. Among phone owners, reported lock use was 50.3% for women (n = 919) and 62.7% for men (n = 1,222). Digital payment use was 34.8% in the lowest income quintile and 67.4% in the highest; corresponding lock use among phone owners was 40.6% and 75.2%. For those aged 60 or older, lock use among phone owners was 22.9% (n = 154), compared with 75.7% for ages 15–24 (n = 497). These measures use different denominators, and the results identify descriptive gaps rather than their causes.

5.3 Adjusted association

The weighted logistic model includes 2,145 phone owners and 219 exposure reports. Holding the included demographic variables constant, digital-payment users had 3.45 times the odds of reporting an unsolicited request compared with non-users (approximate 95% interval 2.24–5.30; independent-observation robust $p < 0.001$). The odds ratio is not a causal effect or a relative risk. Female respondents had lower adjusted odds than male respondents (odds ratio 0.47; approximate interval 0.32–0.69). These intervals do not account for clustering or stratification, so they are exploratory rather than design-based evidence. Other coefficients and complete subgroup counts are in the

accompanying results files.

The original age, income and educational categories, missingness rules, weighted denominators and model computation are documented in the accompanying analysis script. The data remain subject to the World Bank's use and redistribution conditions [2].

5.4 Qualitative interpretation of published sources

Method and scope. This is a focused interpretive review of three public institutional sources, selected for their relevance to trust, consumer protection and fraud prevention: the World Bank discussion of trust [8], its financial consumer protection guidance [9], and the Reserve Bank of India's awareness circular [10]. The sources were read for recurring mechanisms and design implications. This is not a systematic review or an analysis of interviews, survey comments or focus groups; no primary qualitative fieldwork was conducted. The themes below are interpretations, not claims that the survey respondents expressed these views.

Theme 1 Access must become confident use. The World Bank describes trust, awareness and dispute resolution as factors that support adoption [8]. The survey shows high account ownership but lower digital payment use. Together these observations make it reasonable to investigate how trust and usability influence the transition from account access to active use. They do not establish that distrust caused the measured gap: phone access, affordability, preferences and other factors could also matter.

Theme 2 Safety depends on the provider as well as the user. The World Bank emphasizes disclosure, fair treatment and complaint mechanisms as parts of consumer protection [9]. RBI's awareness circular describes recurring fraud methods, including requests for payment credentials and malicious links [10]. In the present data, a device lock is a limited safeguard: it cannot establish whether a user will recognize a deceptive instruction. A useful service should therefore combine understandable guidance with verified support channels and provider-side controls rather than assume a lesson alone will prevent fraud.

Theme 3 Inclusion requires accessible support. The descriptive differences by gender, income and age suggest that a single smartphone-first interface may not fit all users. This is a design hypothesis rather than a finding about individual preferences. Interviews in local languages, testing on inexpensive devices, and observation of how people find help could identify barriers before a product is launched. Participants should not be asked to share credentials or disclose actual account details.

Interpretive synthesis. The quantitative pattern identifies where further investigation is warranted; the literature suggests plausible mechanisms and service features. Neither establishes a market size, a causal effect of a safety tool, or proof that the proposed venture is commercially viable. A pilot should test both adoption and unintended effects, including whether warnings delay legitimate transactions.

6 Entrepreneurial opportunity

6.1 Proposed service

The observed adoption and phone-lock gaps suggest a testable opportunity for a bank or fintech partner to offer a multilingual digital payment safety companion. The proposed service would explain common scam patterns in short audio and visual lessons, use a simple checklist before a user approves a payment, and provide an easy route to the bank's verified support and fraud-reporting channels. It should work on low-cost devices and be tested with first-time users, including women and small merchants. This is a concept based on survey patterns, not evidence of proven demand or effectiveness.

The first customer could be a bank, payment provider or merchant network that pays for onboarding and support; end users could receive the service without an additional fee. A provider partnership could supply verified support links, though willingness to pay and access to distribution remain untested. The service should never request an OTP, UPI PIN, bank password or remote access. Because warnings can be ignored or misunderstood, the provider should retain clear responsibility for its own fraud controls and complaint handling.

6.2 Validation and financial logic

Interview prospective users and partners before building: test whether they recognize a genuine payment request, where they seek help, and what prevents use of existing channels. Then run a small pilot with informed consent and a comparison group if feasible. Track lesson completion, comprehension on unseen scenarios, the proportion finding the correct reporting channel, false alarms, accessibility and support costs. Longer-term outcomes could include verified fraud reports or losses, but only with adequate sample size, consistent observation periods and privacy safeguards. A rise in reports could reflect improved reporting rather than more fraud.

Unit economics require evidence: contribution per institutional customer equals subscription revenue minus onboarding, support, content and infrastructure costs. The service becomes viable only if recurring revenue covers fixed development and compliance costs at a plausible adoption scale. No market size, price, cost or return is asserted here because none has been validated. An independent evaluation should also check whether warnings discourage legitimate transactions or exclude users who rely on assistance.

7 Limitations and future research

The survey is self-reported, cross-sectional and covers adults, not firms specifically. Digital payment use includes receiving payments and does not mean personal use of a mobile banking app. Phone owners are a selected subgroup; reported unsolicited requests may be legitimate, and a device lock is not a measure of resistance to social engineering. We cannot determine whether adoption or security behavior preceded an event. Recall bias, differences in phone activity and unmeasured factors may explain observed associations. The public file supplies weights but not the clustering or stratification identifiers needed for design-based uncertainty; model confidence intervals use an independent-observation approximation and should be treated cautiously.

Future work should obtain design information if available, reproduce design-based variance estimates, and study verified losses with consent and appropriate privacy safeguards. A randomized evaluation of safety interventions could help identify effects, provided the study protects users and measures unintended consequences. Entrepreneur-specific interviews would test whether the adult survey patterns translate into a business need.

8 Conclusion

India’s 2024 Findex survey shows widespread account ownership alongside uneven digital payment use and device protection. Reported unsolicited requests were more common among phone-owning digital-payment users, even after adjustment in an exploratory model, but the association does not show that digital finance caused exposure. Low-friction safety education and support is a plausible entrepreneurial opportunity whose demand, effect and financial viability remain to be tested.

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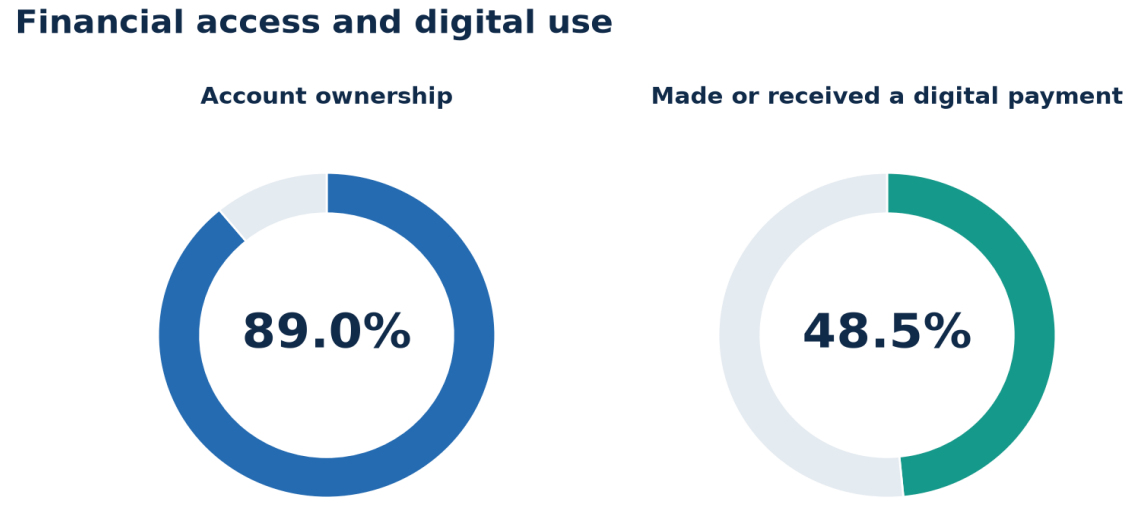
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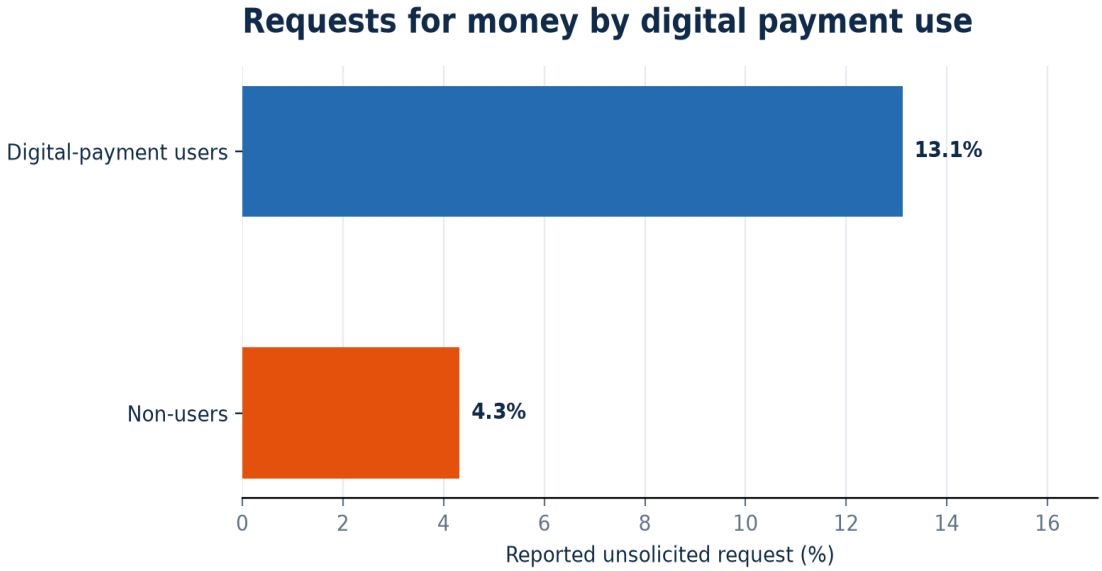
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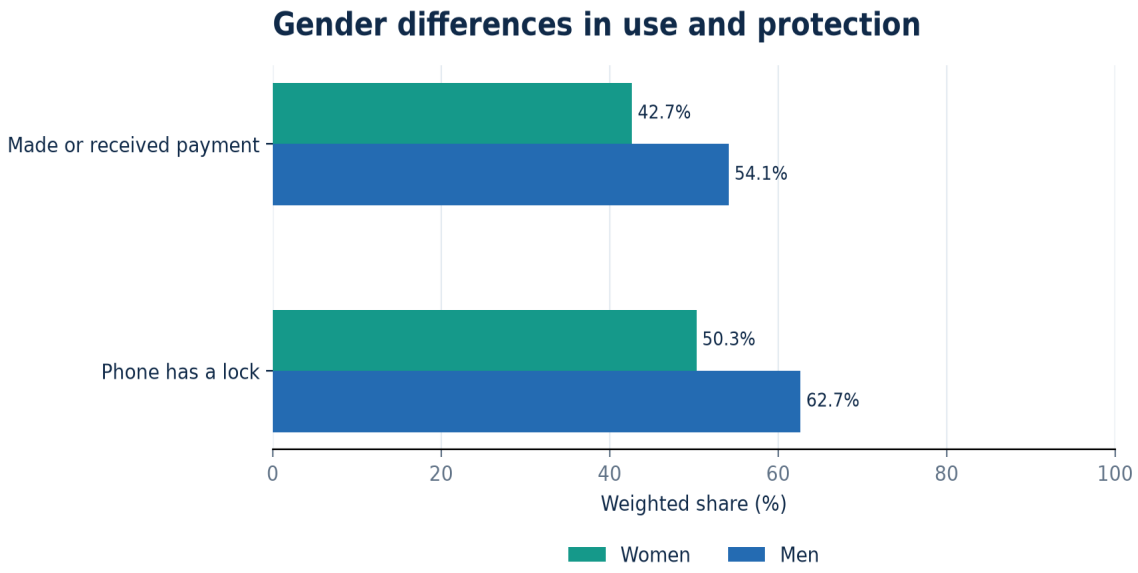
Appendix: Survey figures



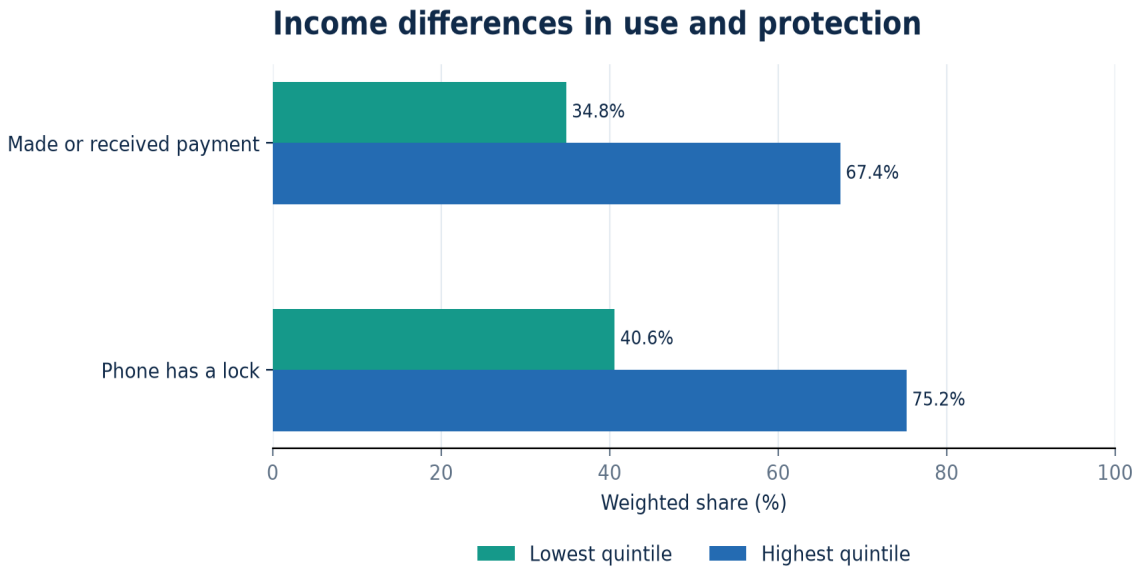
Each ring is a separate share of all adults, n = 3,000. Grey is the remainder.
Source: Global Findex India 2024; Group 31 weighted calculations.



Phone owners: users n = 1,283; non-users n = 862. Requests are not verified fraud.
Source: Global Findex India 2024; Group 31 weighted calculations.



Payment: all adults. Phone lock: valid responses among phone owners.
Source: Global Findex India 2024; Group 31 weighted calculations.



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